Research Statement

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A central challenge for public policy is that markets routinely fail to price externalities—the spillover costs and benefits that economic activity imposes on the environment, on public health, and on other people. My research asks how policies and large shocks reshape those externalities, and who ultimately bears the gains and losses. Across two connected research programs, one centered on the rural economy and one on the urban and regional economy, I use quasi-experimental methods and rich administrative data—including Taiwan's agricultural census, transit records, accident registries, and high-frequency transaction data—to deliver credible causal evidence where it has been scarce. Taiwan offers an unusually informative laboratory: a dense, well-measured economy in which clean-energy programs, transit reforms, and pandemic disruptions arrived as sharp, datable changes that allow before-and-after comparisons of real-world behavior. The two programs share a common goal—turning diffuse, hard-to-see externalities into measured quantities that policymakers can act on.

1. Agricultural, Environmental, and Ecological Economics

Why it matters. Farms sit at the center of two of the defining problems of our time: they must adapt to a changing climate while also helping to cut emissions, and they depend on natural ecosystems whose value rarely shows up in any market price. Governments now intervene heavily—through solar subsidies at home and carbon border tariffs abroad—yet we know surprisingly little about whether these measures actually help the producers and ecosystems they target. Getting this evidence right matters for food security, rural livelihoods, and the credibility of climate policy.

What my research contributes. My core contribution is to show, with farm-level data, that the energy transition and farm profitability need not be in tension. Using Taiwan's comprehensive agricultural census, Liou, Chang, et al. (2025a) finds that installing rooftop solar raises average farm revenue—evidence that energy-transition subsidies can deliver genuine co-benefits rather than merely trading food output for power. A companion study of poultry farms (Lee et al. 2025b) confirms this win–win for both profitability and environmental performance, and adds a policy-relevant nuance: the gains are highly uneven across farm sizes, which matters for how subsidies should be targeted. Moving from the farm to the border, Liou, Chang, et al. (2025b) builds a theoretical framework for the EU's Carbon Border Adjustment Mechanism, clarifying the conditions under which carbon tariffs raise or lower welfare in exporting countries and thereby giving negotiators a clearer map of who wins and who loses. Finally, Lee et al. (2025a) documents that greater forest biodiversity is linked to a more equal distribution of revenue among neighboring farms—novel evidence that ecosystem services act as a natural equalizer for smallholders, and a concrete reason to value biodiversity in economic, not just ecological, terms. In a related study of how cultural practice and the environment interact, Chang et al. (2024) identifies a sharp rise in air pollution around Taiwan's Tomb-Sweeping Day and traces it to holiday traffic rather than the burning of ritual paper—pinning down the public-health cost of a recurring externality and correcting a common misattribution. Together these studies put credible numbers on benefits and costs that climate, health, and conservation debates usually discuss only in the abstract.

2. Urban, Regional, and Transportation Economics

Why it matters. Cities run on mobility, but moving people also generates some of the largest externalities in modern life—traffic deaths, greenhouse gases, and congestion—and the demand shocks that ripple through regional economies can be just as consequential. These effects are notoriously hard to pin down, because transport rules and economic shocks are almost never assigned at random. Credible answers are essential for designing safer streets, cleaner transit, and more resilient local economies.

What my research contributes. I exploit sharp, well-timed policy changes and shocks to recover causal effects that observational studies typically miss. Chang et al. (2021) shows that the steep, lasting drop in Taipei Metro ridership during COVID-19 was driven mainly by people's perception of risk rather than by formal restrictions—an early demonstration that voluntary behavior, not mandates, governed pandemic mobility. Building on how commuters respond to incentives, Liou, Kim, et al. (2025) finds that monthly transit passes in Taiwan's major cities significantly raised ridership and cut carbon emissions, identifying a low-cost lever for greener urban travel. On safety, Liou and Chang (2026b) provides rare causal evidence that removing hook-turn restrictions at busy intersections reduced serious accidents, showing that small design choices at the intersection level carry real safety stakes; Liou and Chang (2026a) complements this by quantifying the elevated accident risk surrounding large-scale religious tourism, turning a recurring event into actionable guidance for traffic management. Extending the lens from mobility to the broader regional economy, Liou et al. (2024) applies a Bayesian change-point model to high-frequency credit card data to trace how Taiwanese consumers shifted their spending during the pandemic, uncovering asymmetric precautionary and substitution patterns that survey data would have missed, while Liou et al. (2026) examines how hotels rebuilt revenue and employment afterward, documenting the role of labor-market rigidity in shaping recovery. Across these studies, my contribution is a body of credible estimates that translate hard-to-identify urban and regional externalities into evidence policymakers can use.

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